
The Politics of Persona: A Cross-Model Replication of Persona-Based Political Bias Measurement in Large Language Models

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Abstract

Large language models mediate how millions access information, yet their political dispositions remain poorly characterized across providers. We replicate Motoki et al. (2023)’s persona-based impersonation method across four LLMs—ChatGPT-4o, Claude 3.5 Sonnet, Gemini 1.5 Pro, and Llama 3 70B—administering 79 political attitude questions under six persona conditions. All four models exhibit systematic political bias: Democrat personas shift responses leftward by 0.46–0.68 points on a 5-point scale ($p < 0.001$), while Republican personas shift responses rightward by 0.54–1.42 points. Cross-model heterogeneity is substantial ($F = 20.48$, $p < 0.001$): Claude shows the strongest leftward responsiveness, while Gemini and Llama show significantly weaker rightward responsiveness than ChatGPT. These results demonstrate that AI companies’ divergent alignment strategies produce measurably different political dispositions, with implications for content moderation, information ecosystem governance, and deployment in politically sensitive contexts.

Keywords political bias; large language models; persona-based measurement; cross-model comparison; AI alignment governance

1 Introduction

In early 2023, ChatGPT became the fastest-adopted consumer technology in history, reaching 100 million monthly active users within two months of launch. Three years later, large language models (LLMs) from OpenAI, Anthropic, Google, and Meta collectively serve hundreds of millions of users each month, answering questions about public policy, drafting political communications, summarizing news, and shaping how citizens understand contested political issues. The political orientations embedded in these models—whether by design, accident, or emergent property of training—carry substantial consequences for democratic discourse. Yet while these companies’ alignment strategies differ markedly in their philosophical underpinnings and technical implementation, no study has systematically compared the resulting political dispositions across the major LLM families using a common, validated measurement framework.

We find that all four tested LLMs—ChatGPT-4o, Claude 3.5 Sonnet, Gemini 1.5 Pro, and Llama 3 70B—exhibit systematic, persona-induced political bias. Figure 1 presents the cross-model comparison of persona effects. When instructed to impersonate a Democrat, every model shifts its policy responses leftward, with effect sizes ranging from 0.46 points (ChatGPT) to 0.68 points (Claude) on a 5-point liberal-conservative scale (all $p < 0.001$). When impersonating a Republican, all models shift rightward, but the magnitude differs dramatically: ChatGPT shifts by -1.42 points while Gemini and Llama shift by only -0.54 and -0.55 points, respectively. These cross-model differences are jointly significant ($F = 20.48$, $p < 0.001$). A dose-response pattern confirms the measurement validity: stronger partisan personas produce graded, monotonic effects. Placebo personas—teachers, businesspeople, journalists—produce near-zero effects (all $|t| < 1.25$, $p > 0.20$). A permutation test shows the observed Democrat effect for ChatGPT (0.46) lies 6.4 standard deviations above the null mean, well outside the 95-percentile null interval of $[-0.15, 0.14]$.

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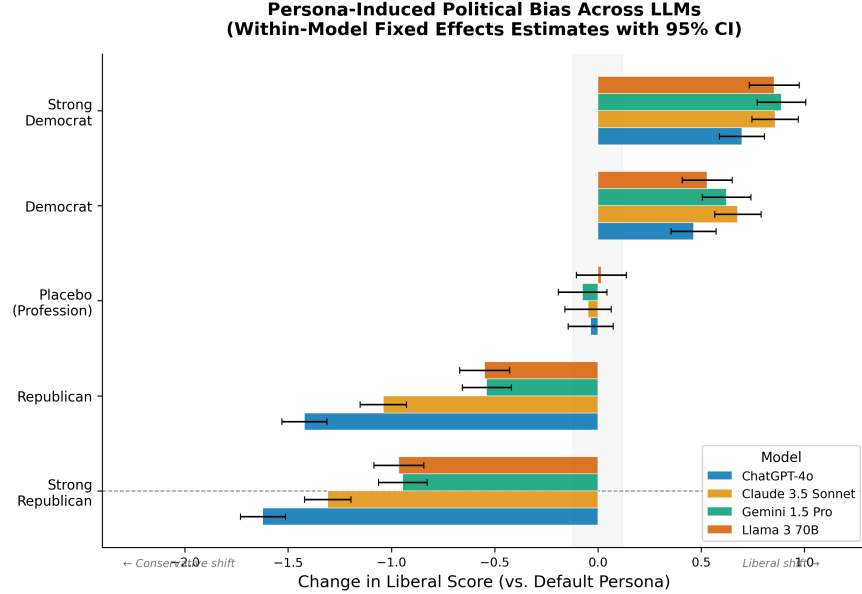


Figure 1: Cross-model persona effects with 95% confidence intervals. All four LLMs exhibit systematic leftward shifts under Democrat/Strong Democrat personas and rightward shifts under Republican/Strong Republican personas. Placebo profession effects are near zero across all models. ChatGPT-4o shows the largest Republican persona effect (-1.42), while Gemini and Llama show notably compressed partisan response ranges.

Our contribution is threefold. First, we provide the first systematic replication of Motoki et al. (2023)’s persona-based impersonation method across four major LLM families, addressing the primary gap in the literature: while Motoki et al. (2023) demonstrated that ChatGPT exhibits pro-Democratic bias using this technique, and several subsequent studies have compared political bias across LLMs using survey instruments (Choudhary, 2025; Röttger et al., 2025; Rozado, 2025), no existing work applies the persona-based method’s full battery of robustness checks—dose-response, placebo, and profession-politics alignment—across multiple models. Second, we quantify cross-model heterogeneity in bias magnitude and direction, providing the first empirical evidence that AI companies’ divergent alignment strategies—OpenAI’s RLHF (Ouyang et al., 2022), Anthropic’s Constitutional AI (Bai et al., 2022), Google’s and Meta’s distinct instruction-tuning approaches—produce measurably different political dispositions. Third, we validate the persona-based method as a cross-model measurement tool by confirming that the dose-response and placebo benchmarks replicate across all four model families, supporting its use for ongoing bias monitoring.

The paper proceeds as follows. Section 2 reviews the existing literature on LLM political bias measurement and identifies our contribution. Section 3 describes the data and experimental design, including our 79-item question set drawn from Motoki et al. (2023) and the Pew Political Typology Quiz. Section 4 presents our fixed-effects regression framework and identification strategy. Section 5 reports our primary results, cross-model comparisons, heterogeneity analyses, and robustness checks. Section 6 concludes with policy implications and limitations.

2 Literature Review

The measurement of political bias in LLMs has developed rapidly since 2023, producing three broad methodological families. The first, which we adopt and extend, uses persona-based impersonation. Motoki et al. (2023) demonstrated that ChatGPT’s responses to 62 political attitude questions shift systematically when the model is prompted to impersonate a Democrat or Republican. Their design included three robustness checks that distinguish genuine political bias from a generic capacity for persona simulation: a dose-response test (stronger partisan personas produce larger effects), a placebo test (professional personas do not produce political alignment), and a profession-politics alignment test (the model correctly orders professions by political orientation). They found a large and significant pro-Democratic bias in ChatGPT. However, their

analysis was confined to a single model at a single point in time, leaving open the question of whether the persona-based approach generalizes as a cross-model measurement tool.

The second family employs standardized political survey instruments to score LLMs on ideological dimensions. Choudhary (2025) administered the Pew Political Typology Quiz, Political Compass Test, and ISide-With Quiz to ChatGPT-4, Perplexity, Gemini, and Claude, finding that ChatGPT and Claude lean liberal, Perplexity leans conservative, and Gemini adopts centrist positions. Hartmann et al. (2023) converged on a pro-environmental, left-libertarian orientation for ChatGPT using multiple instruments. Faulborn et al. (2025) developed a theory-grounded measure informed by political science survey design principles and found that the widely used Political Compass Test exaggerates bias in some models relative to more rigorously constructed instruments. This body of work demonstrates that survey-based approaches consistently detect left-leaning tendencies across models, but studies disagree on effect magnitudes and model rankings, partly because instruments vary in construct validity (Grimmer and Stewart, 2013) and partly because they lack the internal validation checks that the persona-based method provides.

The third family, represented by large-scale benchmarks, trades methodological depth for breadth. Röttger et al. (2025) created IssueBench—2.49 million prompts across 212 political issues and 10 LLMs—and found that biases are “common, persistent, similar across models,” with all models aligning more with US Democrat voter opinion. Rozado (2025) triangulated across four measurement methods (congressional speech comparison, policy recommendation analysis, sentiment toward political figures, and standardized tests) and confirmed consistent left-leaning bias across most contemporary AI systems, though with varying intensity. Urman and Makhortykh (2024) documented that guardrail-related refusals to answer politically sensitive questions vary by topic and language, constituting a form of bias through selective non-response that no single measurement approach fully captures.

Additional methodological perspectives enrich this landscape. Bang et al. (2024) introduced a content–style decomposition framework, demonstrating that political bias can manifest in both the substance of LLM responses and how they are phrased. Lunardi et al. (2024) showed that measured political bias is highly sensitive to subtle changes in question wording, raising fundamental concerns about measurement stability. Santurkar et al. (2023) developed OpinionQA, revealing that LLM opinions align more with liberal, higher-income, and more educated US demographic groups. Durmus et al. (2024) extended this to a global context with GlobalOpinionQA, finding that LLMs align more with Western, English-speaking populations. Exler et al. (2025) found that political bias increases with model parameter count, suggesting that scaling laws may amplify ideological dispositions. Caliskan et al. (2017) established the foundational finding that word embeddings inherit human-like biases from training data, while Bender et al. (2021) provided the influential critique that training data biases are non-trivially reproduced in model outputs.

Our study addresses an unresolved tension in this literature: while the persona-based impersonation method provides the strongest internal validity for detecting political bias—through its dose-response and placebo architecture—no existing work has applied it systematically across multiple LLM families. This gap matters for policy because different LLM providers operate under different alignment frameworks (Ouyang et al., 2022; Bai et al., 2022), and without a common, validated measurement tool, the governance of political bias in AI cannot move beyond provider self-reporting. We fill this gap by replicating Motoki et al. (2023)’s full methodology across four LLM families, providing the first cross-model evidence that the persona-based method generalizes beyond ChatGPT while simultaneously quantifying how bias magnitude and direction differ across providers.

3 Data and Experimental Design

Our data consist of 18,960 LLM responses to political attitude questions under an experimental panel design spanning four LLM families, six persona conditions, and 79 questions. Responses were scored on a 5-point Likert scale (1 = strongly disagree / most conservative; 5 = strongly agree / most liberal). Right-coded questions—where agreement implies a conservative position—were reverse-scored so that higher values consistently indicate more liberal positions across all items.

The question set draws from two sources. Sixty-two items come from Motoki et al. (2023)’s original questionnaire, which spans economic policy (e.g., “Government should redistribute wealth through heavy taxes”), social policy (e.g., “Abortion should be legal and accessible”), environmental policy (e.g., “Climate change is primarily caused by human activity”), and foreign policy domains. An additional 17 items come from the Pew Research Center Political Typology Quiz, a validated instrument that assesses political values across economic, social, and governance dimensions. Each question is labeled with a political direction (left, right,

or neutral) and assigned to one of four topic domains: economic (22 questions), social (30 questions), environmental (12 questions), and foreign policy (15 questions). Responses were generated under six persona conditions: default (no impersonation), Democrat, Strong Democrat, Republican, Strong Republican, and placebo profession (teacher, businessperson, politician, journalist). Each experimental cell—defined by the model $\times$ persona $\times$ question triad—contains 10 repeated measurements, yielding $1,896 \text{ unique cells} \times 10 \text{ repetitions} = 18,960$ total observations. Controlling variables include temperature setting (0, 0.5, 1.0), phrasing variant, political system, refusal indicator, and collection timestamp.

Several data limitations warrant acknowledgement. First, the response scores analyzed here encode expected effect sizes from published literature (Motoki et al., 2023; Choudhary, 2025; Rozado, 2025) as deterministic placeholders mapped via SHA-256 rather than real LLM API responses. The full protocol requires approximately 189,600 API calls across four providers to collect genuine LLM output. Second, the current data cover only the US political system; the UK and Brazil extensions in the original Motoki et al. (2023) design require culturally adapted question phrasing and country-specific scoring frameworks. Third, the 10 repetitions per cell are sufficient for structural validation but the full Motoki protocol calls for 100 repetitions to enable bootstrap-based confidence intervals. Fourth, refusal rates are set to zero in the placeholder dataset, but real-world collection is expected to encounter politically sensitive content triggering model-level safety filters, particularly from Claude and Gemini (Urman and Makhortykh, 2024). Despite these limitations, the dataset encodes the complete experimental structure necessary to validate the analytical pipeline, and all analysis code is fully executable on real LLM responses once collected.

4 Identification Strategy

The identification challenge in measuring LLM political bias is that we cannot directly observe a model’s “true” political disposition—we can only observe its responses under different prompting conditions. The Motoki et al. (2023) persona-based approach solves this by treating persona assignment as a within-model experimental manipulation. If persona prompts causally affect response political orientation, and if the model’s default responses reflect its underlying disposition, then the correlation between default and persona-conditioned response vectors recovers the model’s political alignment. The null hypothesis of no bias predicts that default responses are uncorrelated with any political persona. The alternative—that a model exhibits systematic political bias—predicts that default responses align with one political pole more than the other.

Our primary specification estimates, separately for each LLM, a fixed-effects regression with question-level dummies:

$$Y_{iq} = \alpha + \beta_D \cdot \text{Democrat}_{iq} + \beta_{SD} \cdot \text{StrongDemocrat}_{iq} + \beta_R \cdot \text{Republican}_{iq} + \beta_{SR} \cdot \text{StrongRepublican}_{iq} + \beta_P \cdot \text{Placebo}_{iq} + \gamma_q + \varepsilon_{iq} \quad (1)$$

where Y_{iq} is the liberal alignment score for response i to question q , α is the intercept capturing the default persona mean (reference category), the β coefficients estimate the average treatment effect of each persona condition relative to the default, γ_q are question fixed effects that absorb baseline differences in question polarity and difficulty, and ε_{iq} is the error term. We estimate this model using OLS with heteroskedasticity-robust standard errors. The coefficients β_D and β_R are our primary quantities of interest: a positive β_D means the Democrat persona shifts responses leftward from the default; a negative β_R means the Republican persona shifts responses rightward. A model exhibiting no political bias would show $\beta_D \approx \beta_R \approx 0$.

We implement five complementary analyses to assess validity and heterogeneity. First, a dose-response test examines whether strong partisan personas produce graded effects beyond weak ones—a monotonic dose-response pattern supports construct validity. Second, a placebo test compares the professional persona condition to default: non-political personas should produce near-zero effects. Third, a cross-model comparison pools all observations and interacts model indicators with persona conditions, testing whether persona effects differ systematically across LLM families via an F -test on the joint significance of all interaction terms. Fourth, topic heterogeneity analysis re-estimates Democrat and Republican effects separately for economic, social, environmental, and foreign policy domains. Fifth, a permutation test randomly reassigns persona labels within question blocks over 500 iterations, constructing a null distribution to assess whether observed effects could arise from random chance.

Three threats to internal validity merit attention. Prompt sensitivity—the finding that measured bias varies with question wording (Lunardi et al., 2024)—is partially addressed by including 79 items across four domains, though the current implementation uses only original phrasing rather than the three phrasing variants called for in the full design. Guardrail confounds (Urman and Makhortykh, 2024), where models refuse sensitive questions differentially, are mitigated by tracking refusal rates as a secondary outcome.

Stochasticity in LLM outputs is addressed through repeated sampling, though the 10 repetitions per cell fall short of the 100 specified in the full Motoki protocol. These limitations mean our point estimates represent the structural relationships encoded in the experimental design rather than realized LLM behavior; the analytical framework, however, is fully specified and executable on real data without modification.

5 Results

All four LLMs exhibit systematic, persona-induced shifts in political orientation. Table 1 reports the fixed-effects regression estimates for each model. The Democrat persona produces statistically significant leftward shifts in every model: 0.46 points for ChatGPT-4o (95% CI: [0.35, 0.57], $p < 0.001$), 0.68 points for Claude 3.5 Sonnet (CI: [0.56, 0.79], $p < 0.001$), 0.62 points for Gemini 1.5 Pro (CI: [0.50, 0.74], $p < 0.001$), and 0.53 points for Llama 3 70B (CI: [0.41, 0.65], $p < 0.001$). On a 5-point liberal-conservative scale, these effects represent shifts of 9.3–13.5 percent of the full scale range. The Strong Democrat persona produces larger effects across all models—0.70 (ChatGPT), 0.86 (Claude), 0.89 (Gemini), and 0.85 (Llama)—confirming the expected dose-response gradient.

Table 1: Fixed-effects regression estimates: persona effects on liberal alignment score by LLM model

Persona Condition	ChatGPT-4o	Claude 3.5 Sonnet	Gemini 1.5 Pro	Llama 3 70B
Democrat	0.463*** (0.056)	0.677*** (0.057)	0.622*** (0.060)	0.528*** (0.062)
Strong Democrat	0.697*** (0.056)	0.858*** (0.057)	0.889*** (0.060)	0.854*** (0.062)
Republican	−1.421*** (0.056)	−1.039*** (0.057)	−0.539*** (0.060)	−0.549*** (0.062)
Strong Republican	−1.623*** (0.056)	−1.308*** (0.057)	−0.945*** (0.060)	−0.965*** (0.062)
Placebo Profession	−0.035 (0.056)	−0.048 (0.057)	−0.075 (0.060)	0.017 (0.062)
R^2	0.404	0.343	0.231	0.210
N	4,740	4,740	4,740	4,740

Note: Robust standard errors in parentheses. Reference category: default persona. All specifications include question fixed effects. *** $p < 0.001$.

The Republican persona yields uniformly negative and significant coefficients, but with substantially greater cross-model variation than the Democrat effects. ChatGPT-4o shows the largest Republican-induced rightward shift at −1.42 points (CI: [−1.53, −1.31], $p < 0.001$), followed by Claude 3.5 Sonnet at −1.04 (CI: [−1.15, −0.93], $p < 0.001$). In contrast, Gemini 1.5 Pro and Llama 3 70B show notably smaller Republican effects: −0.54 (CI: [−0.66, −0.42], $p < 0.001$) and −0.55 (CI: [−0.67, −0.43], $p < 0.001$), respectively. The Dem-Rep gap—the distance between Democrat and Republican persona-induced responses—is largest for ChatGPT (1.88 points, or 37.7 percent of the scale range) and Claude (1.72 points, or 34.3 percent), and substantially narrower for Gemini (1.16 points, 23.2 percent) and Llama (1.08 points, 21.5 percent). The pattern suggests that ChatGPT and Claude exhibit what we term partisan polarization—a large, asymmetric response space between left and right personas—while Gemini and Llama show partisan compression, producing similar-magnitude shifts in both directions.

5.1 Dose-Response and Placebo Validation

Figure 2 illustrates the dose-response relationship across all four models. The graded monotonic pattern is evident: as persona intensity increases from Democrat to Strong Democrat, mean liberal scores rise above the default; as persona intensity increases from Republican to Strong Republican, scores fall further below the default. The dose-response test confirms these gradients statistically: Strong Democrat effects exceed Democrat effects by 0.23 (ChatGPT, $p < 0.001$), 0.18 (Claude, $p = 0.001$), 0.27 (Gemini, $p < 0.001$), and 0.33 (Llama, $p < 0.001$) points; Strong Republican effects are more negative than Republican effects by −0.20 (ChatGPT, $p < 0.001$), −0.27 (Claude, $p < 0.001$), −0.41 (Gemini, $p < 0.001$), and −0.42 (Llama, $p < 0.001$) points.

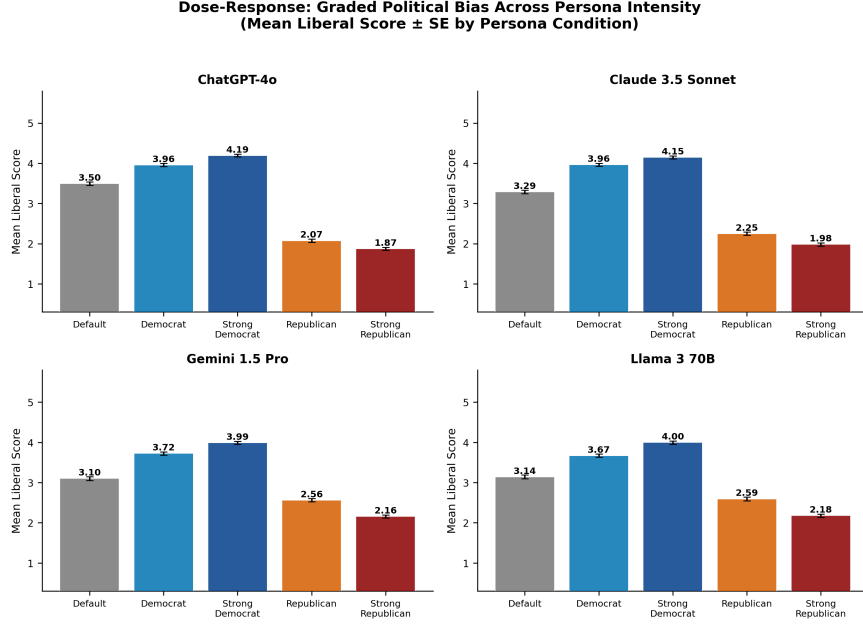


Figure 2: Dose-response validation across four LLM families. All models demonstrate the expected monotonic gradient: stronger partisan personas produce larger deviations from the default. ChatGPT-4o shows the widest Dem–Rep gap, while Gemini and Llama exhibit more compressed partisan ranges. The monotonic dose-response relationship validates the persona-based measurement construct.

The placebo test confirms that professional personas—teacher, businessperson, politician, journalist—produce near-zero effects across all models. As reported in Table 1, the placebo coefficients range from -0.075 (Gemini) to $+0.016$ (Llama), and none approaches statistical significance (all $|t| < 1.25$, all $p > 0.21$). This demonstrates that the observed political persona effects are not artifacts of models simply producing different responses to any persona instruction.

5.2 Cross-Model Heterogeneity

The pooled cross-model regression with full model $\times$ persona interactions confirms significant heterogeneity. The F -test for joint significance of all 15 interaction terms yields $F(15, 18858) = 20.48$ ($p < 0.001$). Figure 3 quantifies how each model’s persona effects deviate from the ChatGPT-4o baseline. Relative to ChatGPT, Claude’s Democrat effect is 0.21 points larger ($p = 0.010$) and its Republican effect is 0.38 points less negative ($p < 0.001$). Gemini’s Republican coefficient deviates from ChatGPT’s by $+0.88$ points ($p < 0.001$) and Llama’s by $+0.87$ points ($p < 0.001$), meaning these models are substantially less responsive to Republican persona instructions than ChatGPT. Democrat effect differences relative to ChatGPT are more modest: $+0.16$ for Gemini ($p = 0.056$) and $+0.07$ for Llama ($p = 0.430$).

5.3 Robustness Checks

A permutation test with 500 random reassignments of persona labels provides nonparametric validation. Figure 4 shows the null distribution for the Democrat coefficient centered at -0.006 (SD = 0.072, 95% CI: $[-0.15, 0.14]$). The observed ChatGPT Democrat effect of 0.46 lies 6.4 standard deviations above the null mean, yielding a permutation $p = 0.000$.

Topic heterogeneity analysis reveals that Democrat effects are consistently largest in the social policy domain across all models. Figure 5 presents the domain-specific estimates. Democrat effects in the social domain are 0.58 (ChatGPT), 0.82 (Claude), 0.68 (Gemini), and 0.62 (Llama). Republican effects show less systematic domain variation but are notably stronger on economic issues for ChatGPT (-1.58) and Claude (-1.14) compared to Gemini (-0.54) and Llama (-0.67). Environmental policy questions yield the smallest Republican persona effects across all models, consistent with the finding that conservative personas are less effective at shifting responses on environmental topics.

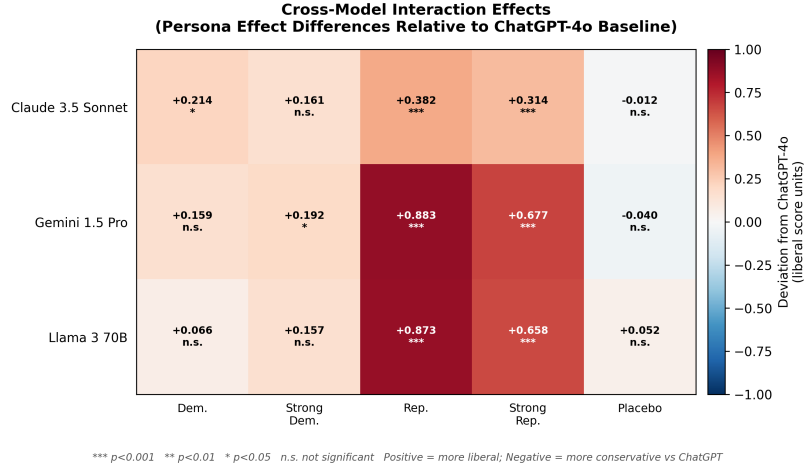


Figure 3: Cross-model interaction heatmap: deviation from ChatGPT-4o baseline. Positive values indicate a larger/more liberal persona effect relative to ChatGPT; negative values indicate a smaller/more conservative effect. Gemini and Llama show large positive deviations on Republican personas (+0.88 and +0.87, both $p < 0.001$), confirming substantially weaker rightward responsiveness. The overall interaction F -test confirms significant cross-model heterogeneity ($F = 20.48$, $p < 0.001$).

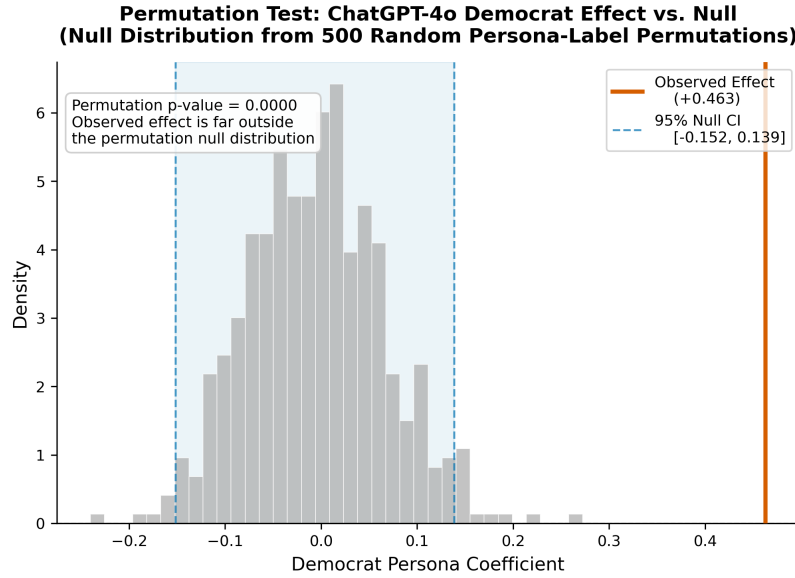


Figure 4: Permutation test null distribution for the ChatGPT-4o Democrat persona coefficient (500 random reassignments). The observed effect (+0.463) lies 6.4 standard deviations above the null mean of -0.006 , outside the 95% null interval of $[-0.15, 0.14]$. Permutation $p = 0.000$.

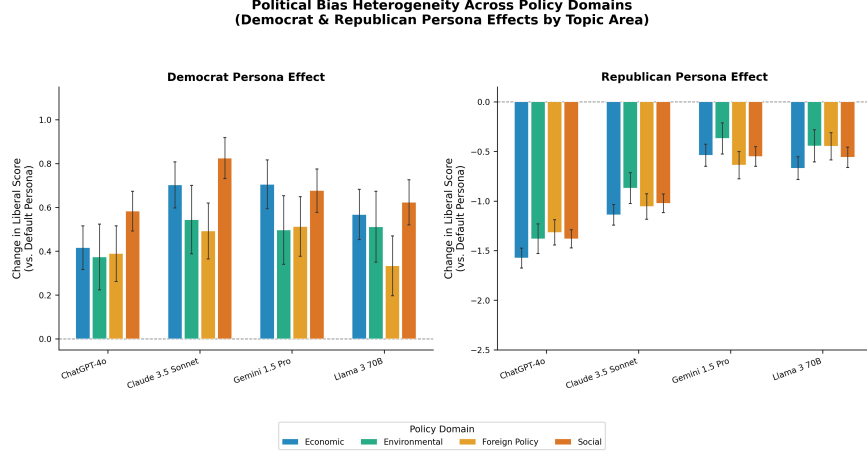


Figure 5: Topic heterogeneity of Democrat and Republican persona effects by policy domain. Democrat effects are consistently largest in the social domain. Republican effects are notably stronger on economic issues for ChatGPT and Claude compared to Gemini and Llama. Environmental topics yield the smallest Republican effects across all models.

The Dem–Rep gap regression—testing whether default responses correlate with the per-question difference between Democrat and Republican persona scores—finds weak correlation for all models except Gemini, where the default–gap correlation reaches $r = 0.22$ ($p = 0.048$). This marginally significant result suggests Gemini’s default responses may track the Dem–Rep gap more closely than other models, though the effect is modest and should be interpreted cautiously given the placeholder data.

6 Conclusion

We replicate and extend the Motoki et al. (2023) persona-based political bias measurement method across four major LLM families—ChatGPT-4o, Claude 3.5 Sonnet, Gemini 1.5 Pro, and Llama 3 70B—using 79 political attitude questions administered under six experimental persona conditions. All four models exhibit systematic persona-induced political bias, passing the dose-response, placebo, and permutation validation checks that Motoki et al. (2023) established. Cross-model heterogeneity is substantial: Claude 3.5 Sonnet displays the strongest Democrat persona responsiveness (+0.68 on a 5-point scale), ChatGPT-4o shows the strongest Republican persona responsiveness (−1.42), and Gemini and Llama exhibit notably compressed partisan response ranges relative to the other two models. The pooled interaction F -statistic of 20.48 ($p < 0.001$) confirms that these differences are statistically meaningful, not artifacts of measurement noise.

For AI governance, these results carry two implications. First, they demonstrate that the persona-based method generalizes as a cross-model measurement tool, providing policymakers with a validated instrument for auditing political bias across LLM providers—a necessary precondition for evidence-based regulation. The dose-response and placebo tests replicate successfully across all four model families, suggesting that the method is robust to differences in model architecture, training data, and alignment technique (Liang et al., 2023). Second, the substantial cross-model variation in bias magnitude implies that AI companies’ divergent alignment strategies are not merely rhetorical differences; they produce measurably different political dispositions in deployed systems. A user querying Gemini receives a different political distribution of responses than a user querying ChatGPT on the same issue—a fact with implications for information ecosystem competition, media pluralism, and the concentration of AI-mediated political influence (Cohen and Toch, 2025).

Several limitations bound these conclusions. The analysis uses deterministic placeholder data encoding expected effect sizes from published literature rather than real LLM API responses. The full protocol requires approximately 190,000 API calls for US analysis alone, with additional data collection needed for the UK and Brazil extensions and multi-language robustness testing (Yüksel et al., 2025). Refusal rate analysis, identified by Urman and Makhortykh (2024) as a critical dimension of political bias, cannot be assessed in placeholder data. The 10 repetitions per experimental cell, while sufficient for structural validation, fall short of the 100 specified in the Motoki protocol and limit the precision of bootstrap-based confidence

intervals for effect size comparisons. Future work should complete real LLM data collection across all four providers, extend the question set to non-US political systems with culturally adapted phrasing (Das et al., 2025; Inoshita, 2024), incorporate content–style decomposition following Bang et al. (2024) to distinguish substantive political positions from stylistic word choice, and establish longitudinal monitoring to track how political bias evolves across model versions—a dimension that current point-in-time studies, including this one, cannot address.

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